1356. Sort Integers by The Number of 1 Bits

Solved

Easy | Topics | Companies | Hint

You are given an integer array `arr`. Sort the integers in the array in ascending order by the number of `1`'s in their binary representation and in case of two or more integers have the same number of `1`'s you have to sort them in ascending order.

Return *the array after sorting it.*

Example 1:

```
Input: arr = [0,1,2,3,4,5,6,7,8]
Output: [0,1,2,4,8,3,5,6,7]
Explantion: [0] is the only integer with 0 bits.
[1,2,4,8] all have 1 bit.
[3,5,6] have 2 bits.
[7] has 3 bits.
The sorted array by bits is [0,1,2,4,8,3,5,6,7]
```

Example 2:

```
Input: arr = [1024,512,256,128,64,32,16,8,4,2,1]
Output: [1,2,4,8,16,32,64,128,256,512,1024]
Explantion: All integers have 1 bit in the binary
representation, you should just sort them in ascending order.
```

Constraints:

- `1 <= arr.length <= 500`
- $0 <= arr[i] <= 10^4$

Python:

class Solution:

```
    def sortByBits(self, arr: List[int]) -> List[int]:
        return sorted(arr, key= lambda a: (a.bit_count(), a))
```

JavaScript:

```
var sortByBits = function(arr) {
    const countBits = (x) => {
        let c = 0;
        while(x > 0){
            if(x & 1) c++;
            x >>= 1;
        }
        return c;
    }

    arr.sort((a,b)=>{
        let bitsA = countBits(a);
        let bitsB = countBits(b);
        if(bitsA === bitsB) return a - b;
        return bitsA - bitsB;
    });

    return arr;
};
```

Java:

```
class Solution {
    public int hammingCode(int n){
        int cnt=0;
        while(n>0){
            n&=(n-1);
            cnt++;
        }return cnt;
    }
    public int[] sortByBits(int[] arr) {
        List<List<Integer>>bucket=new ArrayList<>();
        for(int i=0;i<32;i++){
            bucket.add(new ArrayList<>());
        }
        for(int x:arr){
            int bits=hammingCode(x);
            bucket.get(bits).add(x);
        }
        for(List<Integer>list:bucket){
            Collections.sort(list);
        }
```

```
        int ind=0;
        for(int i=0;i<32;i++){
            for(int num:bucket.get(i)){
                arr[ind++]=num;
            }
        }return arr;
    }
}
```